Do subjective and EEG carryover differences repeat across sessions?

A fixed-catalogue, internally validated analysis of HABS recordings

Research report | 9 November 2026 | Exploratory secondary analysis

Abstract

Objective. Test whether the influence of preceding reported arousal differs between current arousal and posterior EEG alpha, and whether that personal contrast repeats across two sessions. **Methods.** We reconstructed actual video predecessors before exclusions and retained 53 cohort-2 identifiers with 4,554 novel-current-video transitions. Context-adjusted personal slopes were analyzed with their full sampling covariance in a two-session REML model, controlling session length and J1 normalization scales. **Results.** Cross-day random-effect covariance was 0.0035, model-based 95% bootstrap interval [-0.0103, 0.0151]; null-bootstrap p=0.253. The equal-person J2 MAE gain from a fixed J1 personal slope was -0.0055, 95% interval [-0.0123, 9.38e-04]. **Interpretation.** Positive cross-session repeatability of the personal contrast was not established. The primary grouped J2 comparison did not establish a predictive improvement. These findings concern reported arousal and one EEG summary, not clinical stress or the existence of neural correlates of experience.

1. Motivation

Emotional history can modify feelings and peripheral responses differently [2]. Some EEG responses track the current event, whereas post-emotion microstate or aperiodic changes have also been reported [3,4]. A group effect does not establish a reliable personal characteristic [5,6]. We therefore tested the repeatability of a specific history-effect contrast, rather than searching for emotion classes or a universal stress biomarker. Stimulus identity can also confound decoding of assigned emotion categories [8].

2. Data and measurements

Released HABS annotations were reconciled to recording events and clocks [1]. Cohort 2 contained 113 rated identifiers and 9,726 matched trials. Consecutive pairs were defined on the complete video inventory before rating or EEG exclusions. Both videos required valid arousal and CURATED EEG, disjoint ten-second windows, the same recording and a nonnegative gap no longer than 180 s. Current videos previously encountered by that person were excluded. At least twenty remaining pairs per day were required; the final panel contains 2,294 J1 and 2,260 J2 pairs.

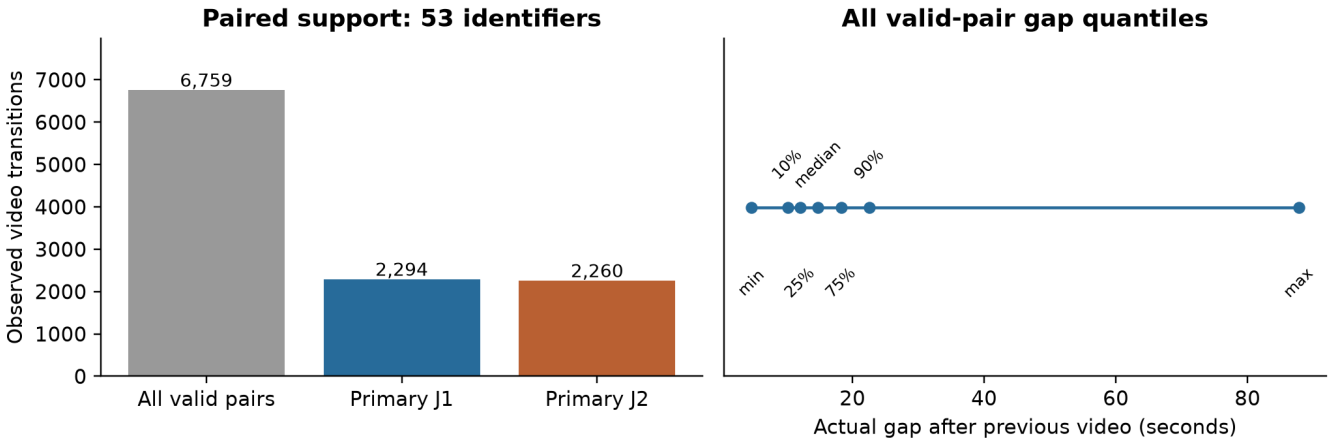


Figure 1. Actual eligible transitions and recorded inter-video gaps. Counts of all valid pairs precede paired-person and novelty restrictions; the gap panel describes that broader valid-pair set. The analysis unit for repeatability is the person, not the transition.

3. Estimand and uncertainty

A is reported arousal. P is minus the logarithm of posterior relative alpha power, 8-13 Hz relative to 1-40 Hz, summarized from usable P3/P4/O1/O2 channels. The existing extraction also requires at least two usable frontal channels. Each measurement is centered and scaled with that person's observed J1 reference. P is an operational EEG summary; its sign does not make it a quantity of stress.

Each day was analyzed separately. Fixed nuisance terms included current and previous video, person intercept, pre-existing expected-emotion/source descriptors, order, onset, their quadratic terms, centered log gap and interactions of gap with both preceding measurements. The reference gap was 15.957 s, the J1 median. Each person had separate preceding-arousal and preceding-EEG slopes. The contrast delta is the preceding-arousal coefficient for current A minus that for current P.

Linear projection retained the full covariance of the two response systems. Sampling covariance for all personal contrasts retained dependence induced by shared nuisance fitting, with an actual-event-order Bartlett kernel through lag two and HC3 leverage correction. The resulting two-day estimates were analyzed with correlated sampling errors and an unstructured two-session random-effect covariance [7]. Day-specific mean terms controlled log pair counts and the known J1 arousal/EEG scale ratio. This ratio is essential: a shared normalization can otherwise resemble stable personal variation.

Variance components used REML, multiple optimization starts, 999 fitted-model bootstrap draws for covariance uncertainty and 999 zero-covariance null draws for the likelihood-ratio reference. Correlation was profiled and not interpreted when weakly identified. These intervals condition on estimated sampling covariance and normal random effects; they do not remove error in lagged ratings or EEG.

4. Identifiability and pre-effect precision

Both days identified all 106 personal-history columns. Nuisance ranks were 579 and 584; history condition numbers were 5.17 and 19.13. Before fitting empirical effects, synthetic slopes were generated under the real predictor geometry. Small between-person variation produced broad correlation estimates, while larger variation was more precisely recovered. These simulations diagnose this estimator and design under specified noise assumptions, not empirical effects or guaranteed power.

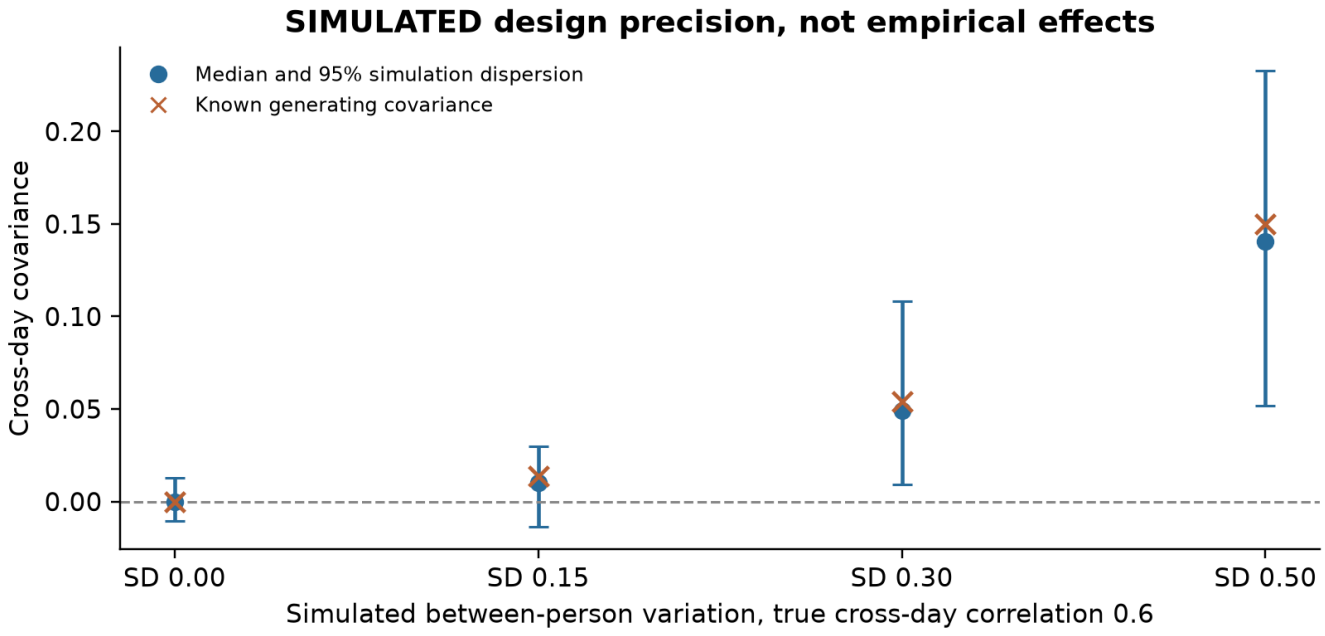


Figure 2. SIMULATION. Known generating covariance and median/95% dispersion across 100 synthetic replicates per scenario, using the observed design and assumed serial noise. A separate shared-scale-only scenario produced a raw correlation around 0.56 with zero generating residual personal covariance; the adjusted model centered covariance near zero.

5. Cross-session contrast results

Quantity	Estimate
Paired identifiers / transitions	53 / 4,554
Cross-day random-effect covariance	0.0035
Model-based 95% bootstrap interval	[-0.0103, 0.0151]
Two-sided null-bootstrap p	0.253
Between-person SD, J1 / J2	0.0450 / 0.0781
Approximate correlation profile range	[-0.9950, 0.9951]

Positive cross-session repeatability of the personal contrast was not established. The latent correlation was weakly identified; its point estimate is not treated as evidence of stability. A mean subjective-EEG slope difference and a reproducible personal difference are separate hypotheses. The scatter is descriptive; its raw correlation is not the primary repeatability estimate.

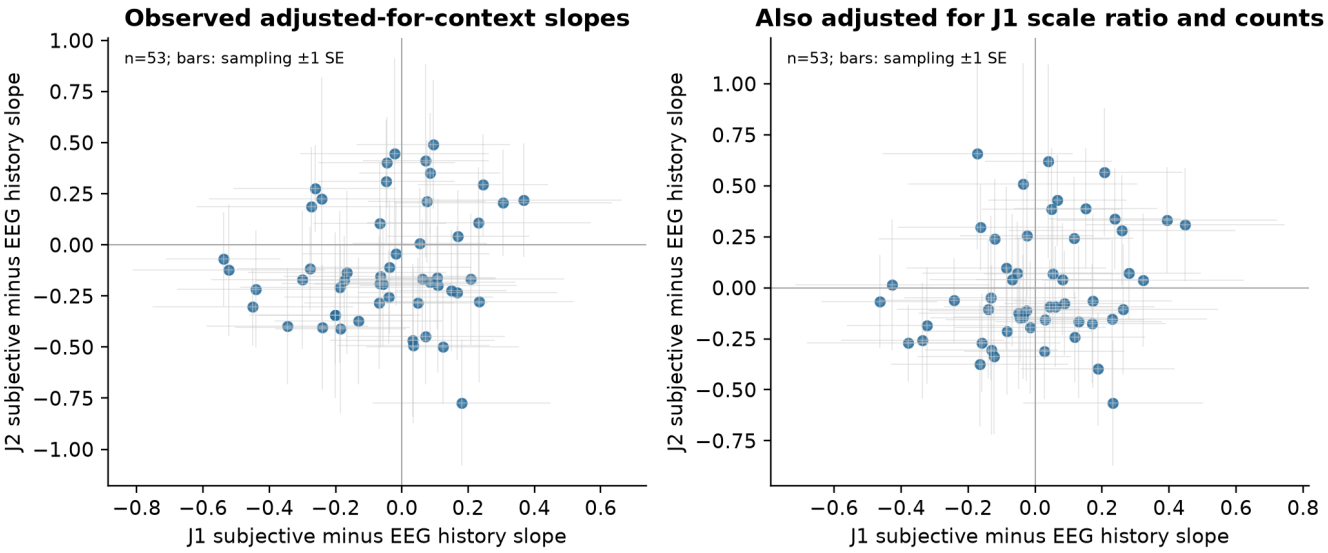


Figure 3. Actual personal slope contrasts with sampling-error bars. The right panel additionally removes the fitted day-specific count and scale-ratio mean terms. These are noisy estimates; neither panel alone establishes latent personal stability.

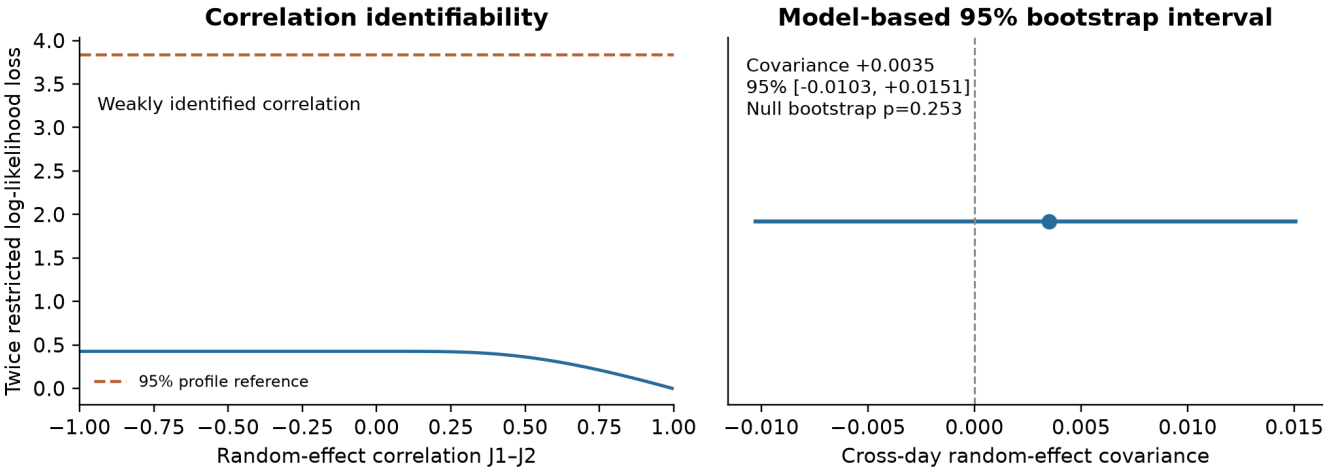


Figure 4. Restricted-likelihood correlation profile and the conditional covariance bootstrap interval. The horizontal profile reference is approximate, especially near a variance boundary. A boundary point estimate is not evidence of near-perfect repeatability.

6. Predictive corroboration on J2

We predicted current standardized D=A-P in five participant-grouped folds, using only current pre-existing context, already observed predecessor A/P and the predicted person's J1 calibration. No current response was a predictor. Population transforms used other training IDs only. All models used population ridge alpha=10 and personal slope penalty=20, fixed before scientific fits. The primary comparison removed only the personal lagged-A correction while retaining the same fitted intercept and lagged-P correction. A separately refitted restricted calibration was an additional reference.

Frozen predictor	Equal-person MAE
Exact personal-slope ablation	1.2124
Full personal carryover correction	1.2179
Refitted restricted calibration	1.2138

The primary gain was -0.0055, participant-bootstrap 95% interval [-0.0123, 9.38e-04]; 21/53 identifiers improved. Against the refitted restriction, gain was -0.0041, interval [-0.0110, 0.0025]. Positive gain means lower error. The primary grouped J2 comparison did not establish a predictive improvement. The prediction grid comprised 2,260 J2 transitions.

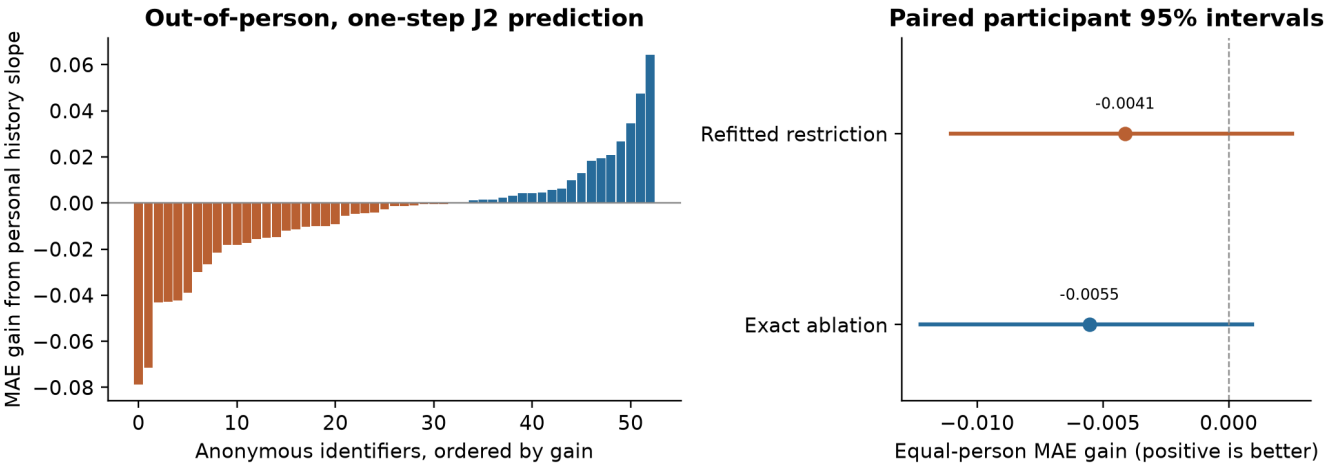


Figure 5. Equal-person out-of-fold gains from adding the personal J1 history slope. Intervals resample identifiers 10,000 times. Previously reported J2 responses are allowed only as one-step observed predecessors; this is not a prediction of the entire future session without updates.

7. Interpretation

The study distinguishes temporal dependence in a measured response from a stable personal attribute and from a useful forecast correction. An apparent relationship between noisy J1/J2 slopes can be supported by stimulus composition, shared calibration or session length. The specified controls and error model narrow the claim, while predictive validation asks a separate practical question. Sensitivities are reported without replacing the primary result by a more favorable specification.

8. Limits and reproducibility

This is an exploratory secondary analysis after the previous HABS study's reserved outcomes were opened. J2 is an internal later-session check, not an independent new cohort or preregistered confirmation. The fixed catalogue and novel-to-person rule do not demonstrate generalization to entirely new videos. Cross-cohort biological identities were not linked.

Ratings occur after each clip and are not continuous reports of experience. There is no no-rating manipulation to separate emotional persistence from response habits. A single scalp-alpha summary, short windows and heuristic artifact screens cannot characterize all neural information. Sampling covariance is estimated; lagged-variable error, J1-reference uncertainty, random-effect nonnormality and residual temporal confounding remain. Common-scale, count, gap and covariance sensitivities delimit robustness rather than prove causality.

The findings do not establish a clinical stress measure, an interoceptive deficit, a biological participant type or a mechanism of qualia. A failure to establish repeatability of this contrast would not imply absence of neural correlates of subjective experience. Any positive personal association requires independent confirmation.

Reproducibility

Settings and source/input fingerprints were frozen before real scientific fits. The native cache was verified against its original extraction manifest. Both real-data cold-grid smoke tests were required. Skore forecasts and the saved covariance likelihood were audited read-only, with original private bindings restored and without estimator fitting. Source snapshot: 5b079a836da5. Public evidence and the executable study accompany this report; individual trial records remain in team storage.

References

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[8] Gerster M et al. Stimulus identity rather than emotion drives EEG classification on the FACED dataset. bioRxiv, 2026, preprint. doi:10.64898/2026.06.12.731889.